IC-GN Initial-Guess Strategy Evaluation

7 strategies × 15 scenarios × 19 frames each

Total IC-GN + init time: 53.9s

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Goal
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Quantify how much the choice of initial guess affects IC-GN
accuracy, convergence, and runtime across diverse motion
scenarios (constant velocity, acceleration, reversal,
impulse, chirp, stop-and-go, random, plus spatial
discontinuities).

Strategies
----------
  previous        - U_t = U_{t-1}  (current default)
  linear_extrap   - U_t = 2 U_{t-1} - U_{t-2}
  narrow_fft_R5   - per-node FFT, ±5 px window around
                    U_{t-1}, auto-expands on clipped peaks
  adaptive_t{N}   - 'previous' with FFT fallback when the
                    previous IC-GN convergence rate fell
                    below N%
  fft_every       - full FFT every frame (baseline)

All FFT paths use the pyALDIC auto-expand loop (2× retries up
to image/2).
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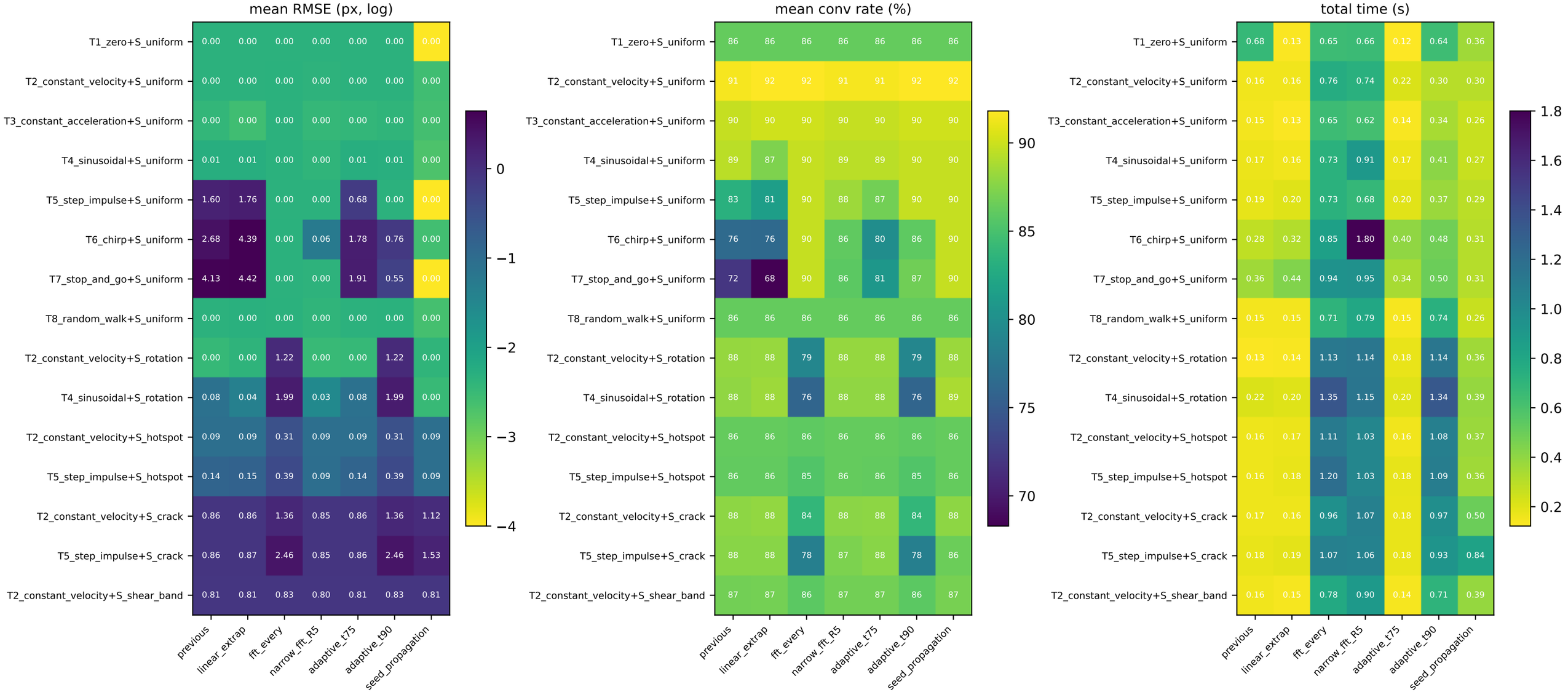
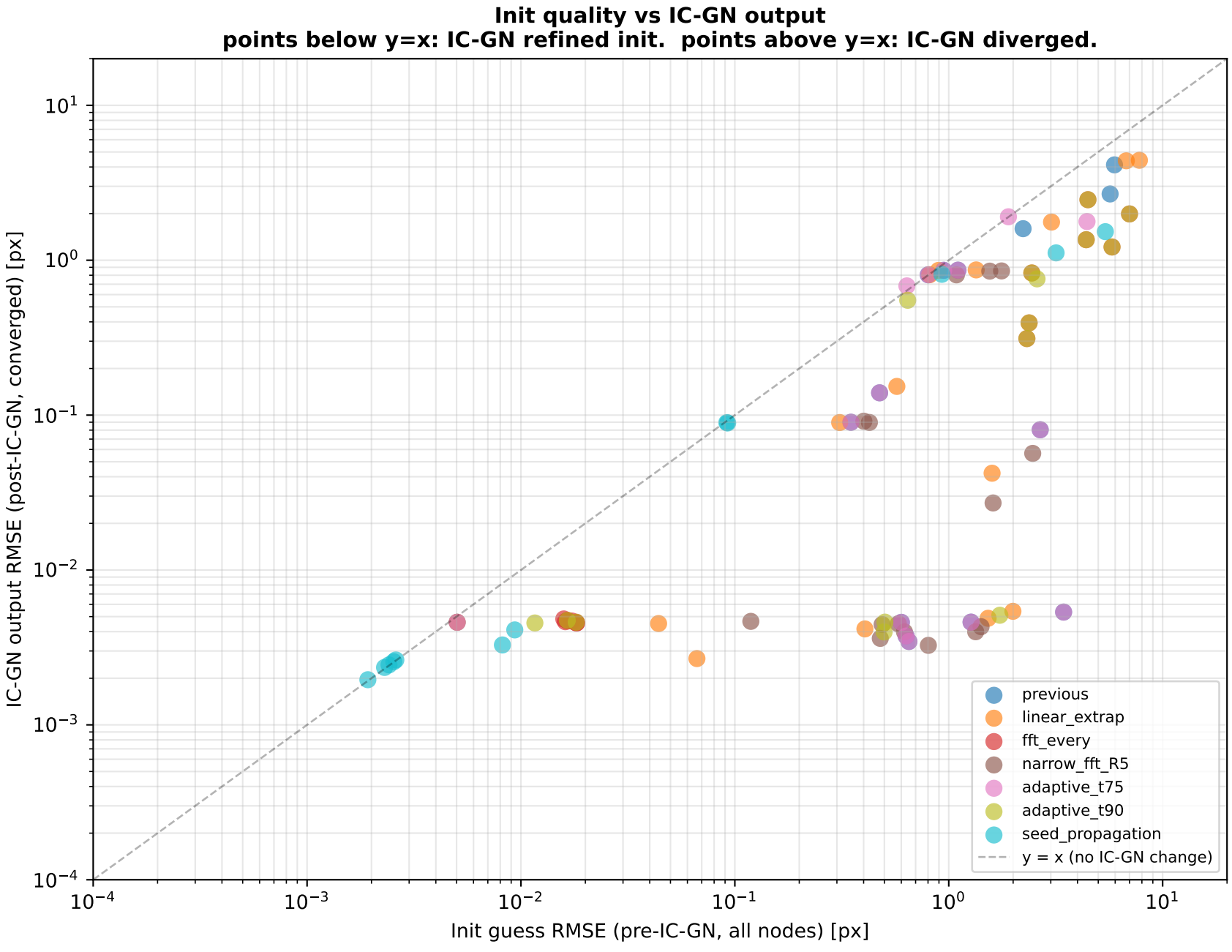
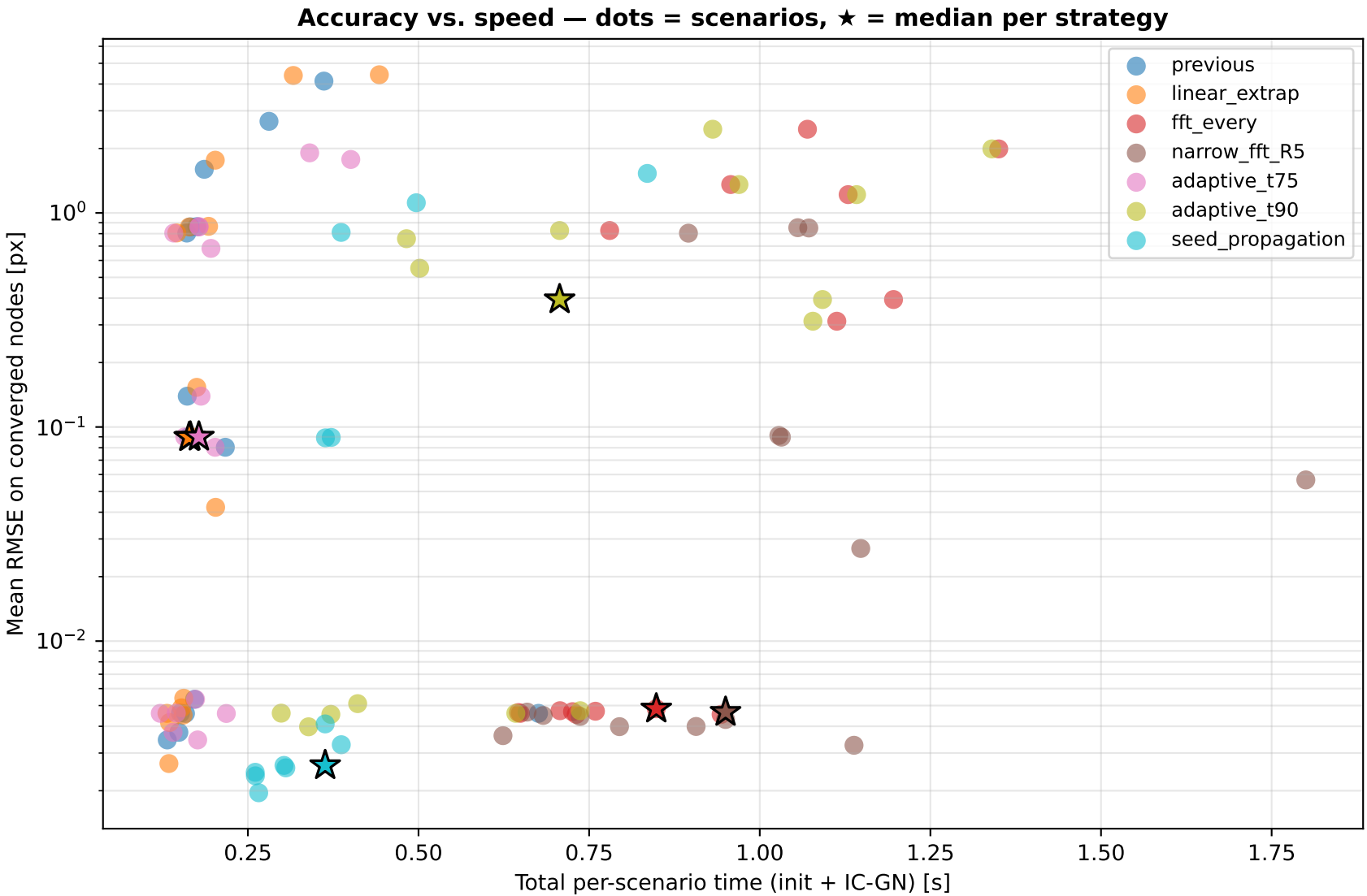
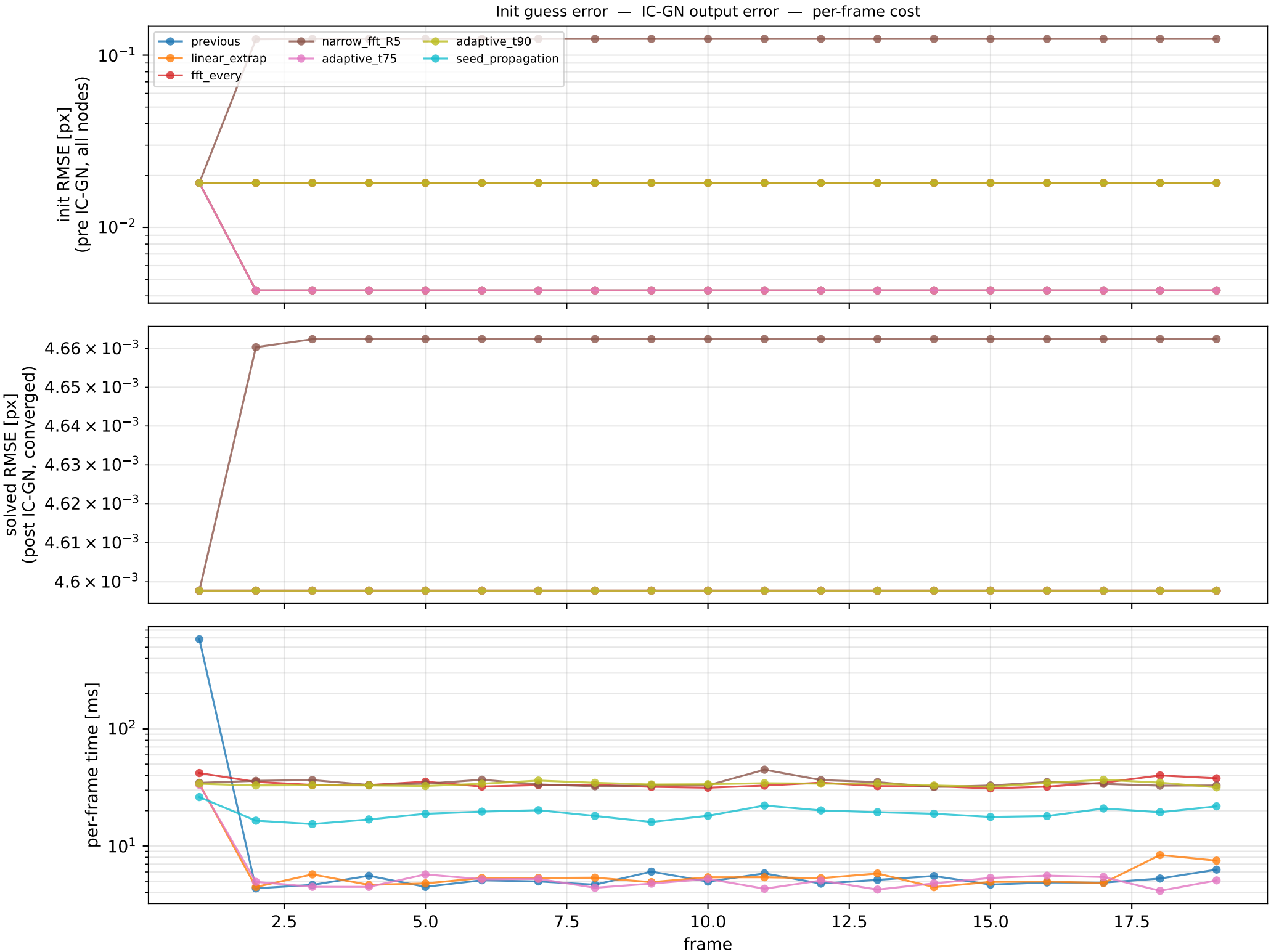
Per-(strategy, scenario) summary




Init quality vs IC-GN output
points below y=x: IC-GN refined init. points above y=x: IC-GN diverged.
IC-GN output RMSE (post-IC-GN, converged) [px]
Init guess RMSE (pre-IC-GN, all nodes) [px]
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
previous
linear_extrap
fft_every
narrow_fft_R5
adaptive_t75
adaptive_t90
seed_propagation
y = x (no IC-GN change)


Accuracy vs. speed — dots = scenarios, ★ = median per strategy
previous
linear_extrap
fft_every
narrow_fft_R5
adaptive_t75
adaptive_t90
seed_propagation
Mean RMSE on converged nodes [px]
10^{0}
10^{-1}
10^{-2}
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Total per-scenario time (init + IC-GN) [s]

T1_zero+S_uniform



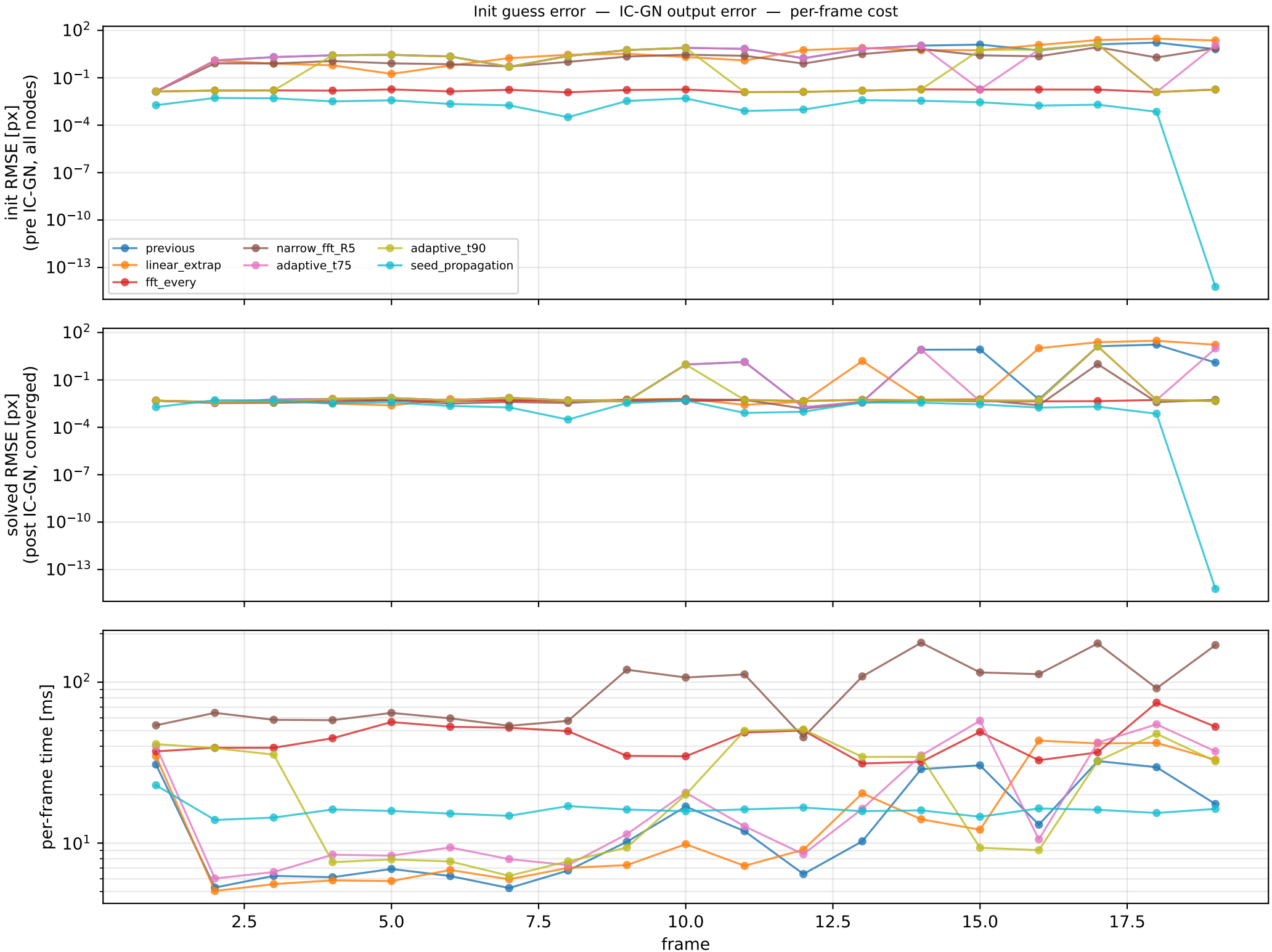
T2_constant_velocity+S_uniform

T3_constant_acceleration+S_uniform

T4_sinusoidal+S_uniform

T5_step_impulse+S_uniform

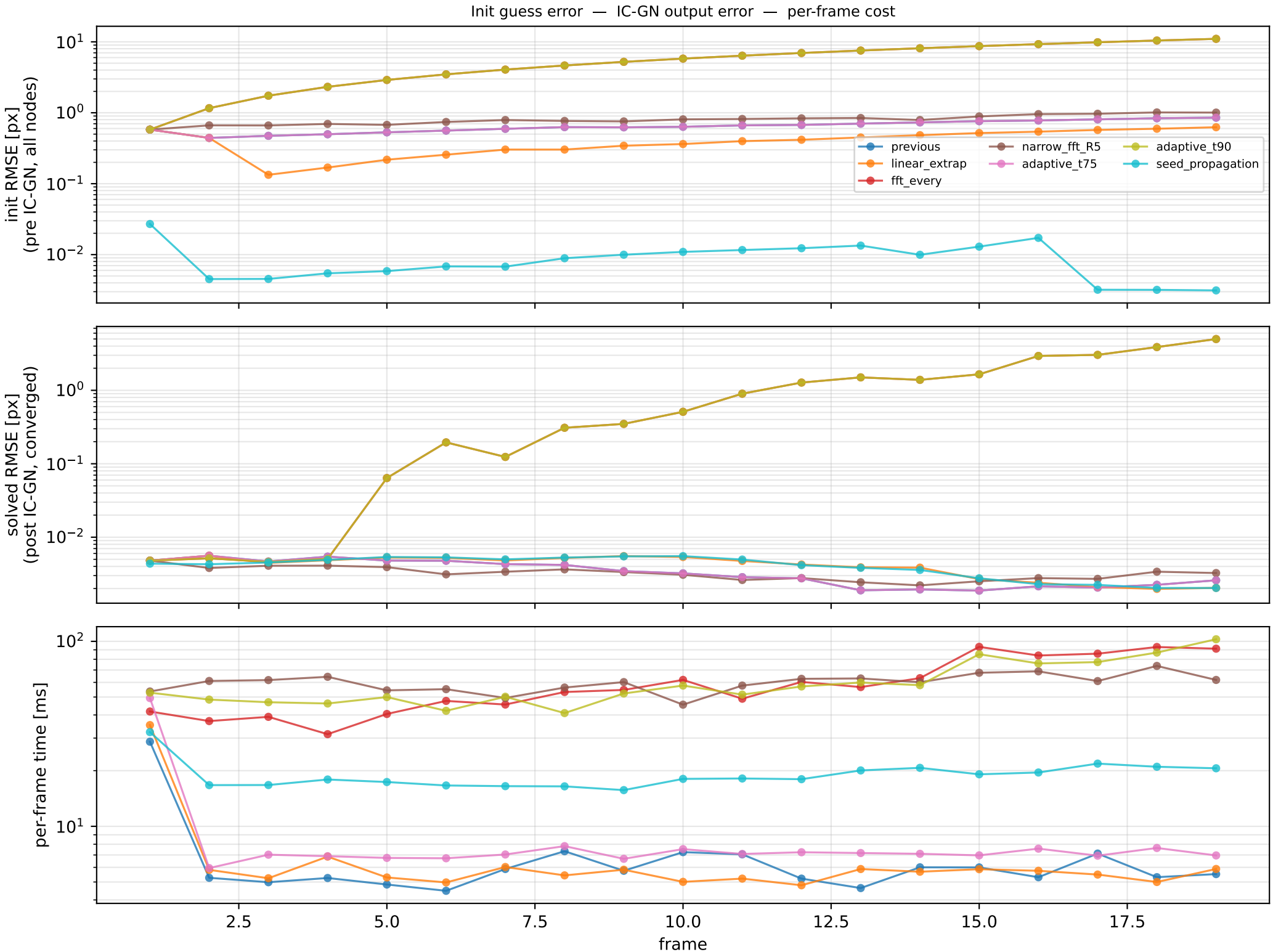
T6_chirp+S_uniform



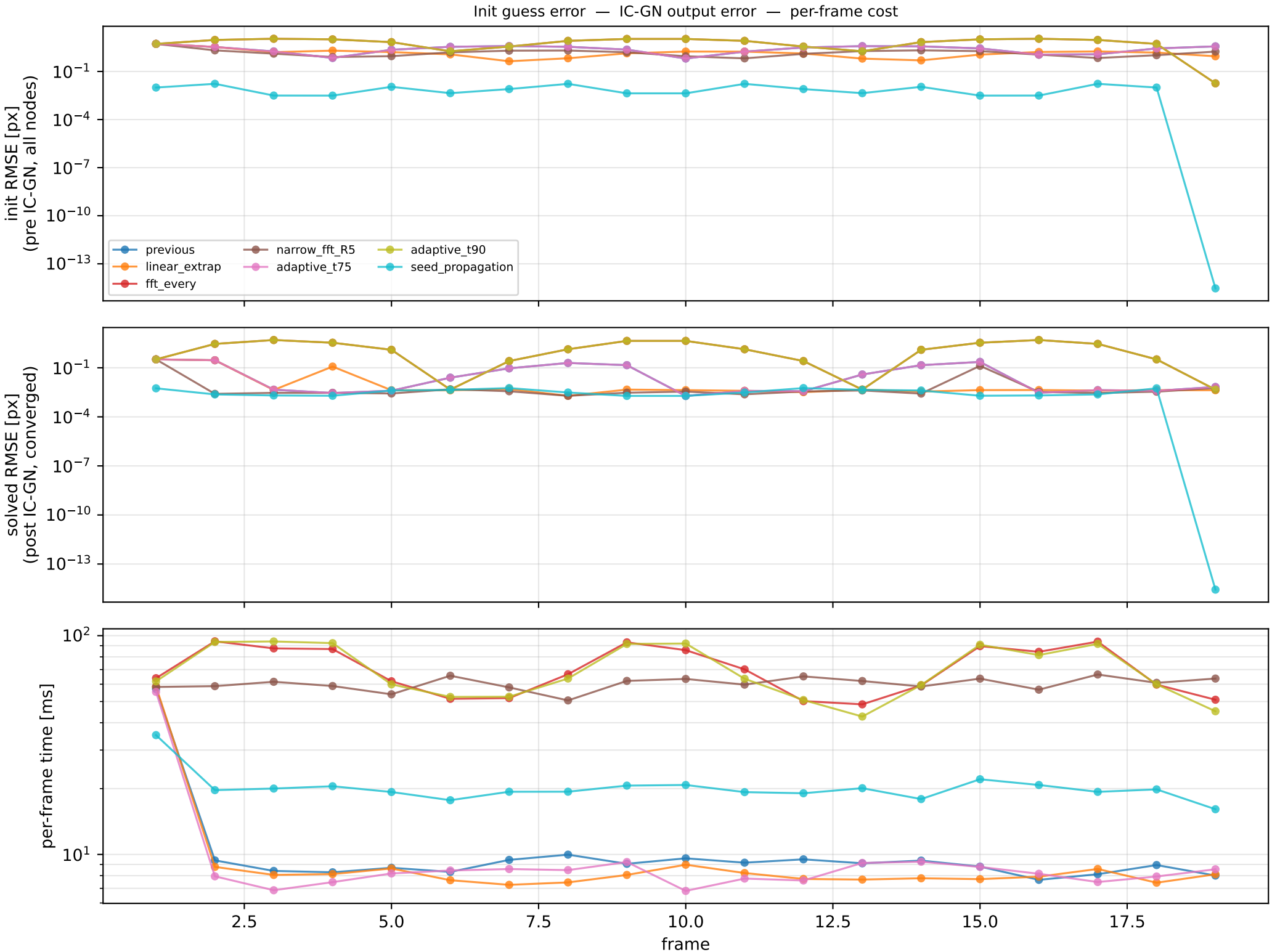
T7_stop_and_go+S_uniform

T8_random_walk+S_uniform

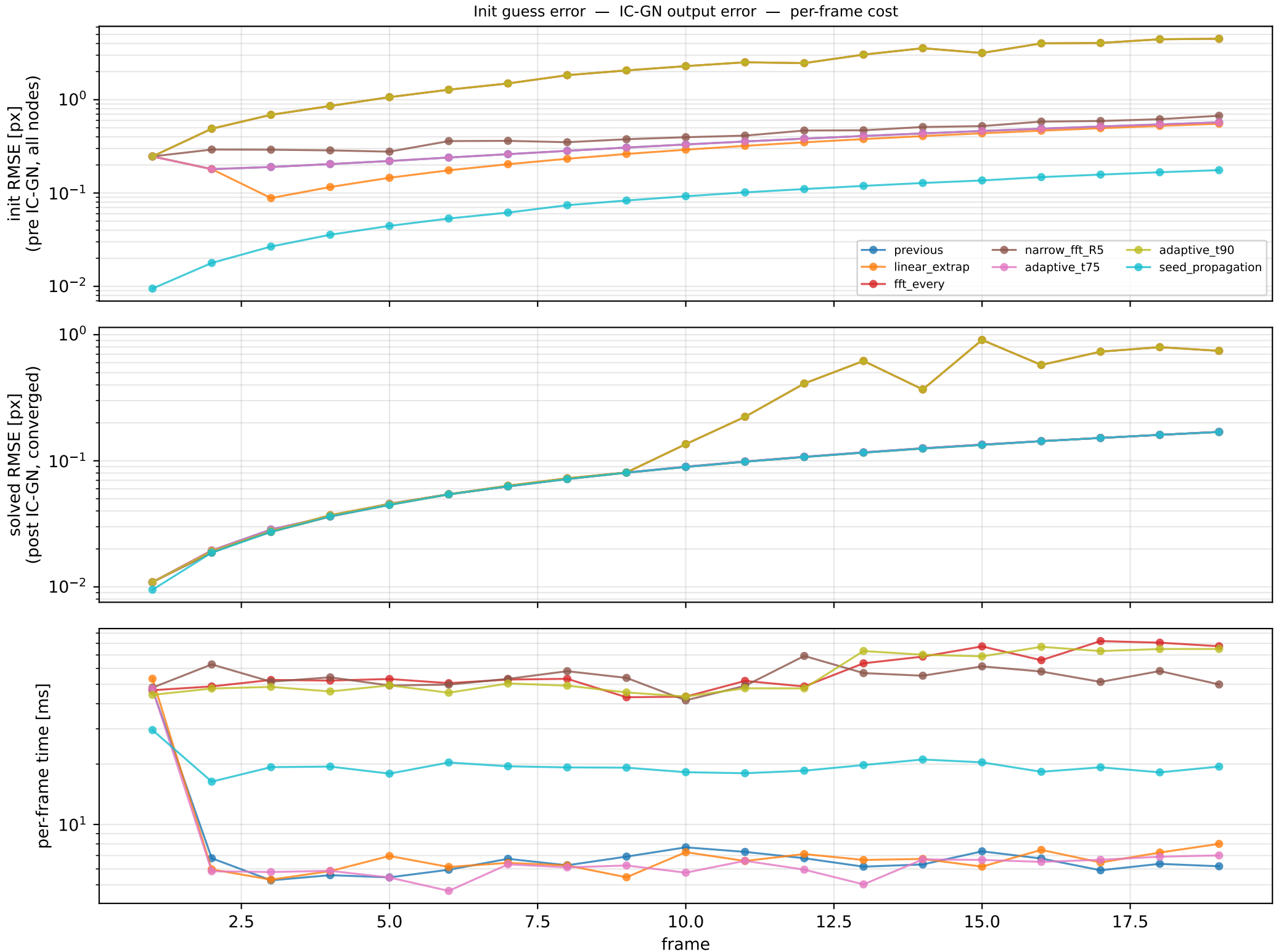
T2_constant_velocity+S_rotation



T4_sinusoidal+S_rotation

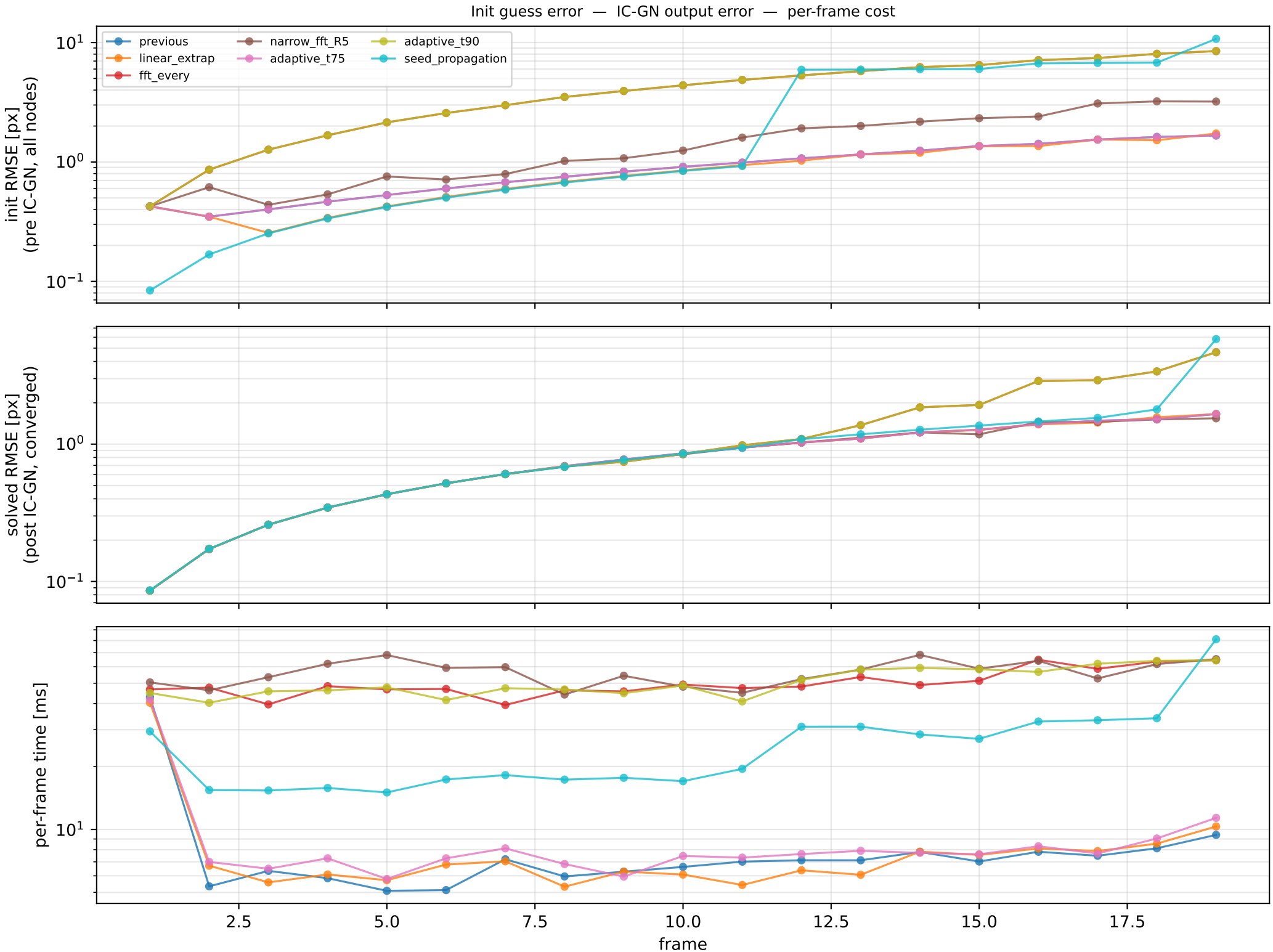


T2_constant_velocity+S_hotspot



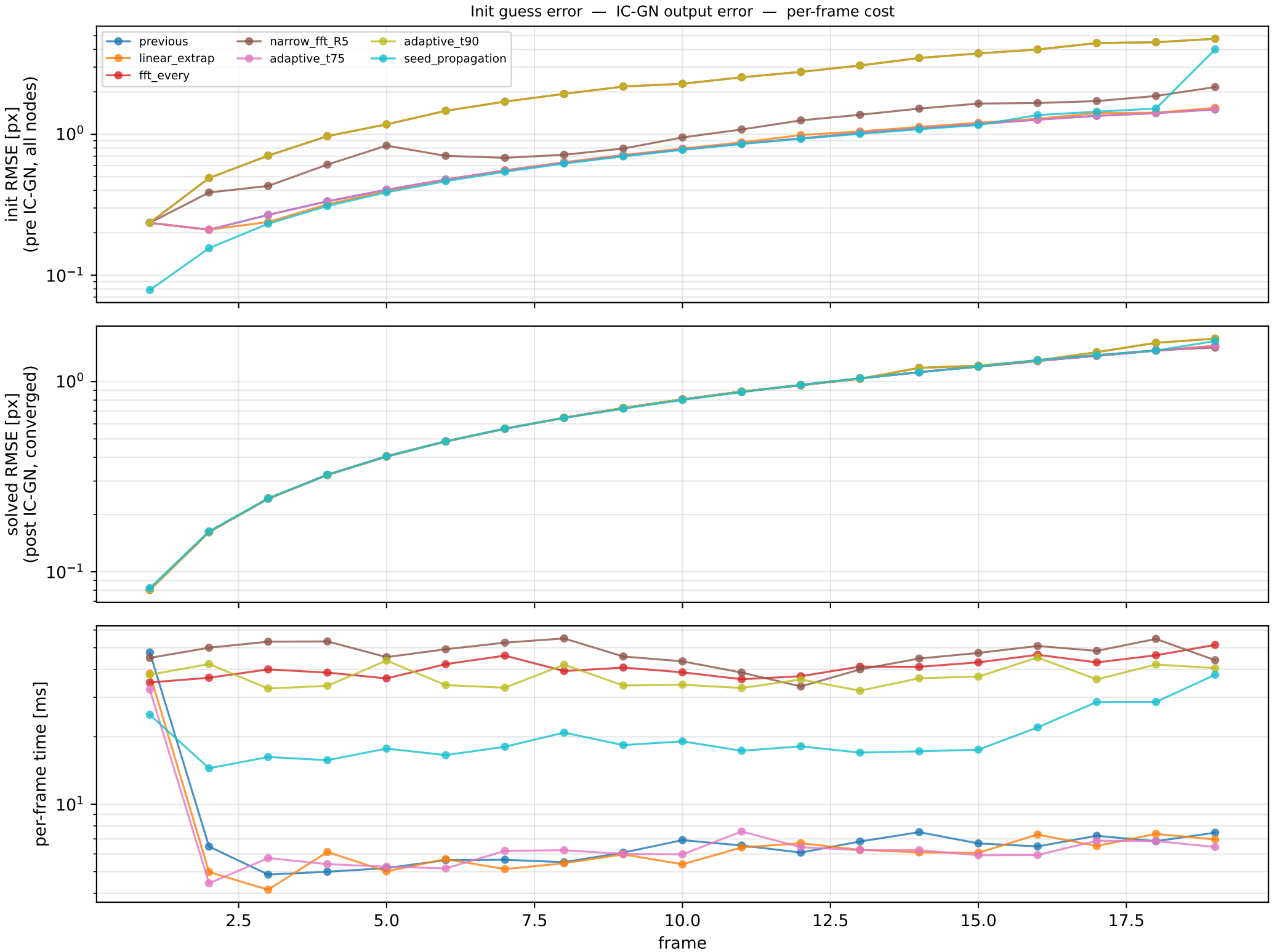
T5_step_impulse+S_hotspot

T2_constant_velocity+S_crack



T5_step_impulse+S_crack

T2_constant_velocity+S_shear_band



Scenario → recommended strategy

Scenario	Best strategy	RMSE (px)	Conv %	Time (s)	Notes
T1_zero+S_uniform	seed_propagation	0.000	86	0.36	1.8× faster than fft_every
T2_constant_velocity+	seed_propagation	0.003	92	0.30	2.5× faster than fft_every
T3_constant_accelerati	seed_propagation	0.002	90	0.26	2.5× faster than fft_every
T4_sinusoidal+S_unifor	seed_propagation	0.002	90	0.27	2.7× faster than fft_every
T5_step_impulse+S_un	seed_propagation	0.000	90	0.29	2.5× faster than fft_every
T6_chirp+S_uniform	seed_propagation	0.003	90	0.31	2.8× faster than fft_every
T7_stop_and_go+S_uni	seed_propagation	0.000	90	0.31	3.1× faster than fft_every
T8_random_walk+S_ur	seed_propagation	0.002	86	0.26	2.7× faster than fft_every
T2_constant_velocity+	narrow_fft_R5	0.003	88	1.14	
T4_sinusoidal+S_rotati	seed_propagation	0.003	89	0.39	3.5× faster than fft_every
T2_constant_velocity+	seed_propagation	0.089	86	0.37	3.0× faster than fft_every
T5_step_impulse+S_ho	seed_propagation	0.089	86	0.36	3.3× faster than fft_every
T2_constant_velocity+	narrow_fft_R5	0.850	88	1.07	
T5_step_impulse+S_cra	narrow_fft_R5	0.853	87	1.06	
T2_constant_velocity+	narrow_fft_R5	0.804	87	0.90	